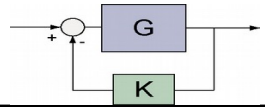


ME 2801

Introduction to Control Systems

W3: Learning to think in the s-domain



So far we have...

- Modeled simple (lumped-parameter) systems
- Introduced some MATLAB/Simulink
- Introduced the Laplace Transform for solving LTI, ODEs

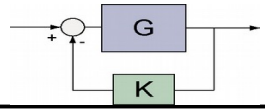
Which means...

- We can make Transfer Functions of mechanical systems
- We can use MATLAB to manipulate those Transfer Functions

This week...

- We'll work on tools for examining the performance (speed, overshoot, steady-state error) and stability of systems based on solely on their transfer function representations

Views of System Building Blocks



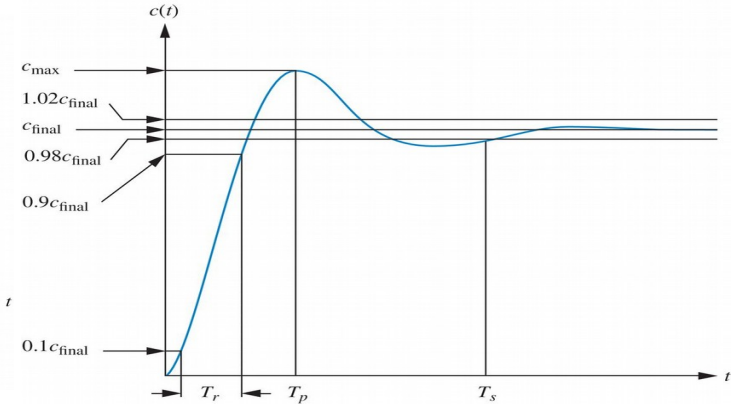
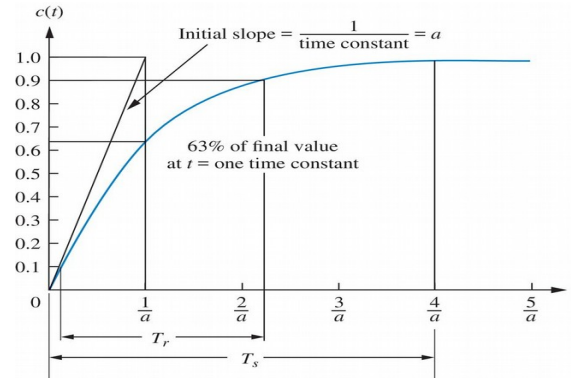
LTI ODEs:

$$M\dot{u}(t) + Du(t) = T(t)$$

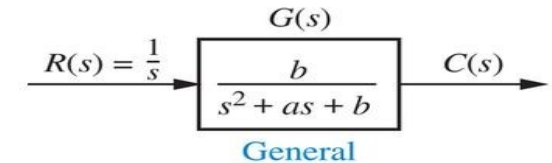
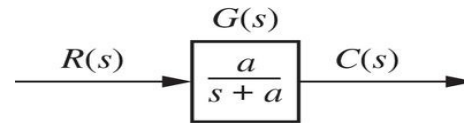
$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(t)$$

Time-Response

- (step, ramp, sinusoid, etc.)

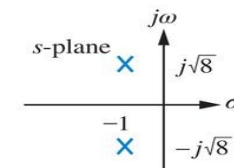
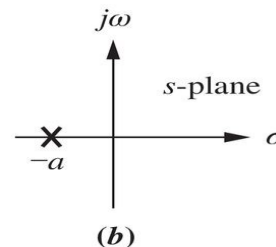


Transfer Function

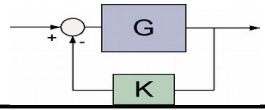


s-Plane

(poles and zeros) So far we have...



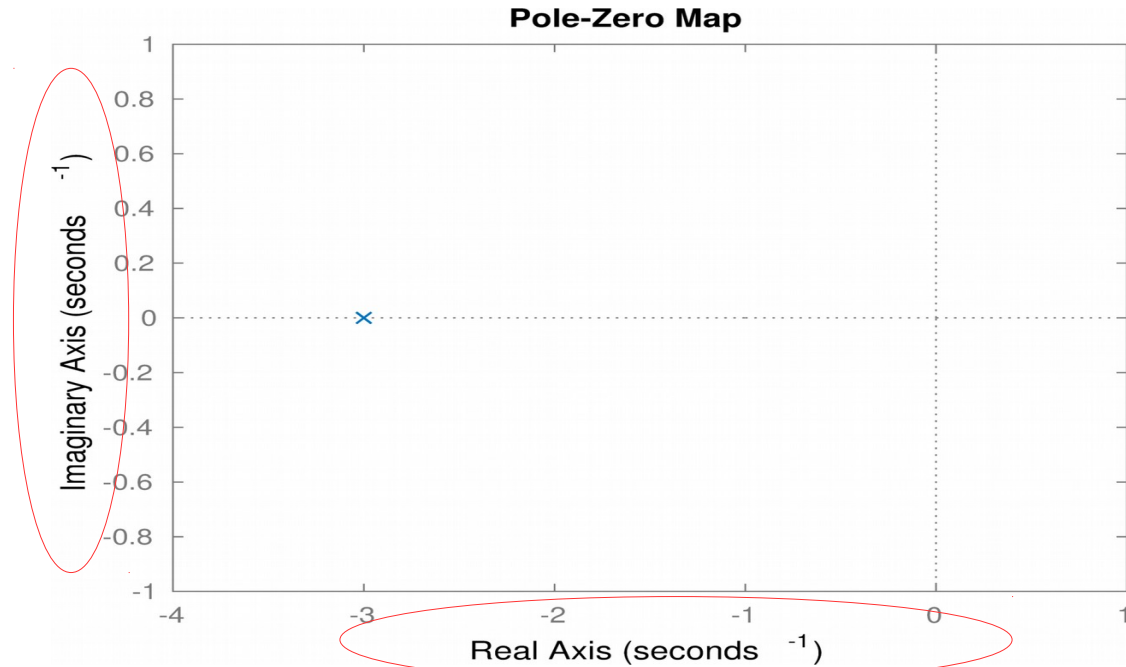
Objects to Think With



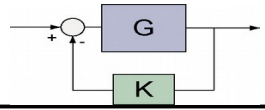
Poles = roots of the denominator = roots of the characteristic equation

Location of system poles is the key indicator of system performance

- Much of classical control is the process of putting these poles in “the right” place.
- Poles tell us...
 - Is the system stable
 - How fast the system responds
 - Will there be steady-state error
 - Will it oscillate and if so how much



Time-Constant: Impulse Response



First-order system

- DC Gain: K
- Time constant: $\tau = 1/a$

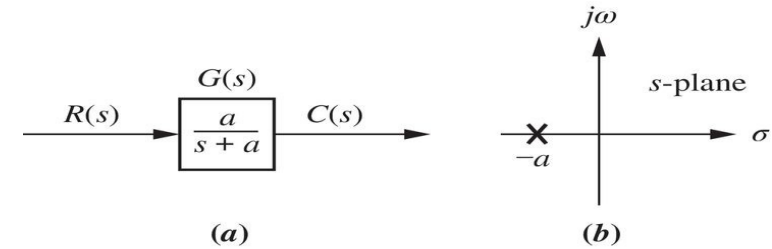
$$G(s) = K \frac{a}{s+a} = K \frac{1}{\tau s + 1}$$

- Impulse Response

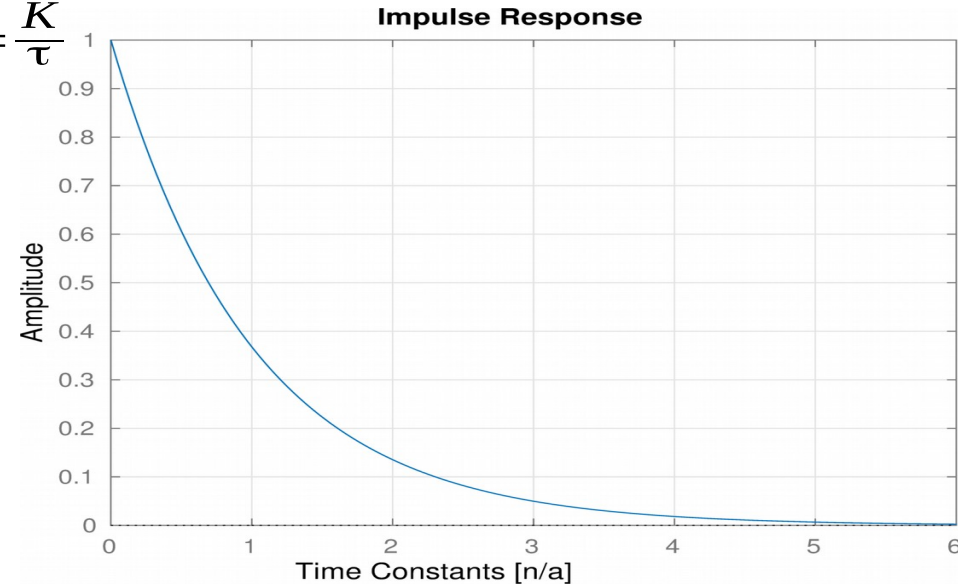
$$C(s) = R(s)G(s) = K \frac{a}{s+a} = K \frac{1}{\tau s + 1}$$

$$c(t) = K a e^{-t/\tau} = \frac{K}{\tau} e^{-t/\tau}$$

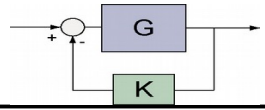
t	$c(t) = \frac{K}{\tau} e^{-t/\tau}$
$1(\tau)$	$c(t) = \frac{K}{\tau} e^{-1} = 0.368 \left(\frac{K}{\tau} \right)$
$2(\tau)$	$c(t) = \frac{K}{\tau} e^{-2} = 0.135 \left(\frac{K}{\tau} \right)$
$3(\tau)$	$c(t) = \frac{K}{\tau} e^{-3} = 0.050 \left(\frac{K}{\tau} \right)$



$$K a = \frac{K}{\tau}$$



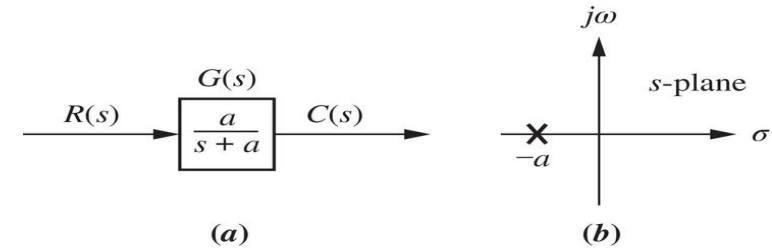
Time-Constant: Step Response



First-order system

- DC Gain: K
- Time constant: $\tau = 1/a$

$$G(s) = K \frac{a}{s+a} = K \frac{1}{\tau s + 1}$$

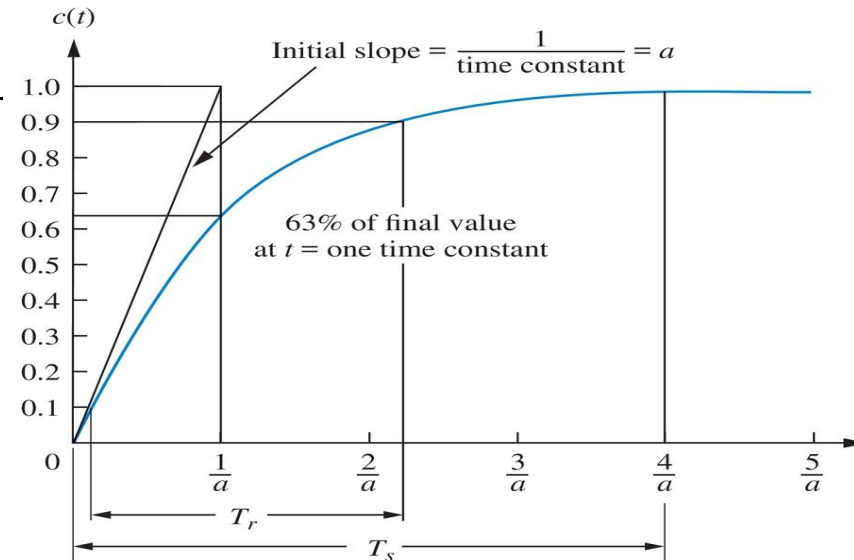


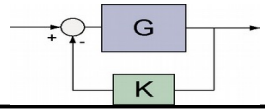
- Step Response

$$C(s) = R(s)G(s) = K \frac{a}{s(s+a)} = K \frac{1}{s(\tau s + 1)}$$

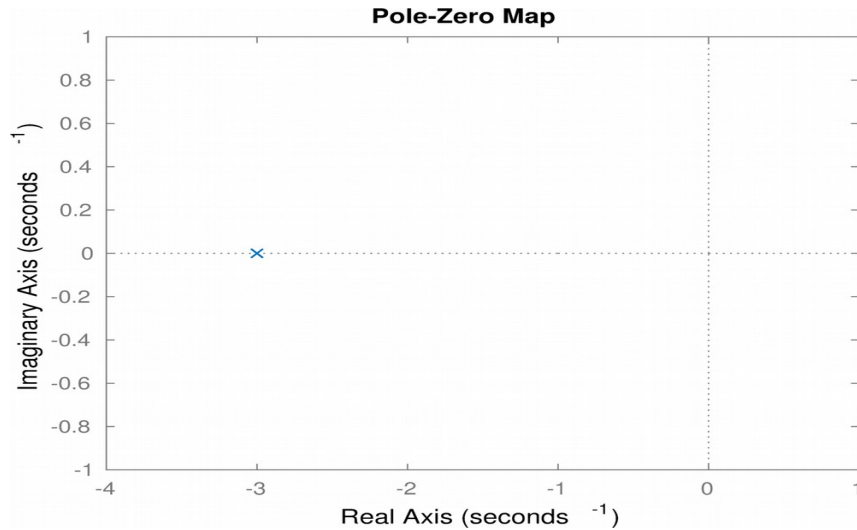
$$c(t) = K(1 - e^{-t/\tau})$$

t	$c(t) = K(1 - e^{-t/\tau})$
$1(\tau)$	$c(t) = K(1 - e^{-1}) = 0.632(K)$
$2(\tau)$	$c(t) = K(1 - e^{-2}) = 0.865(K)$
$3(\tau)$	$c(t) = K(1 - e^{-3}) = 0.950(K)$

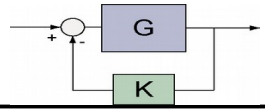




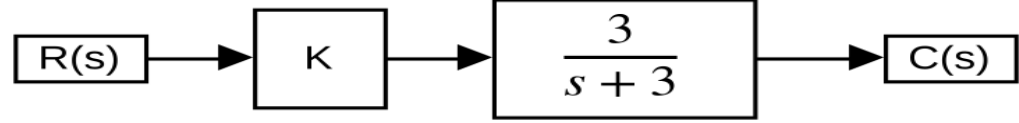
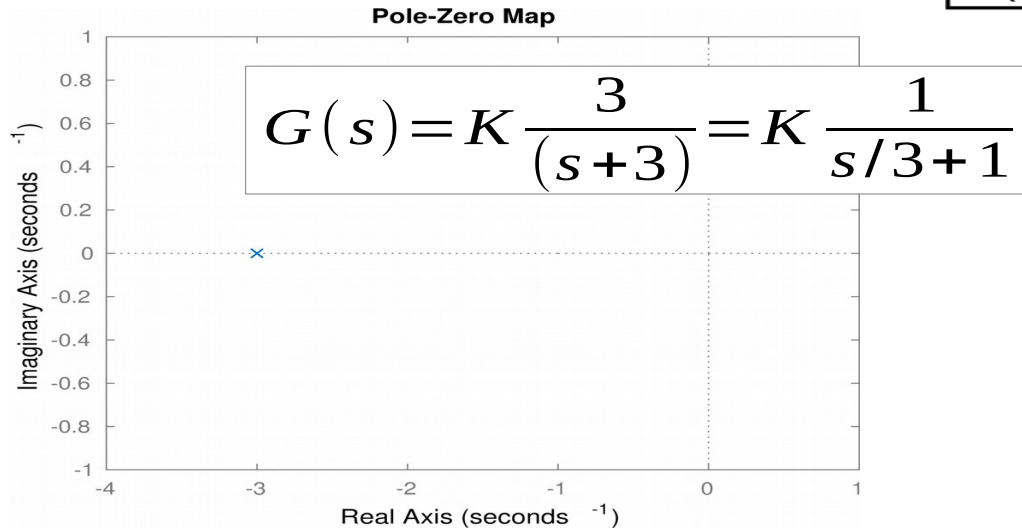
Given this picture



- Write the TF
- Evaluate stability
- Sketch the impulse response
- Sketch the step response
- Find the time-constant

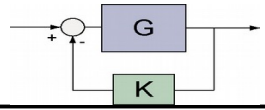


Given this picture

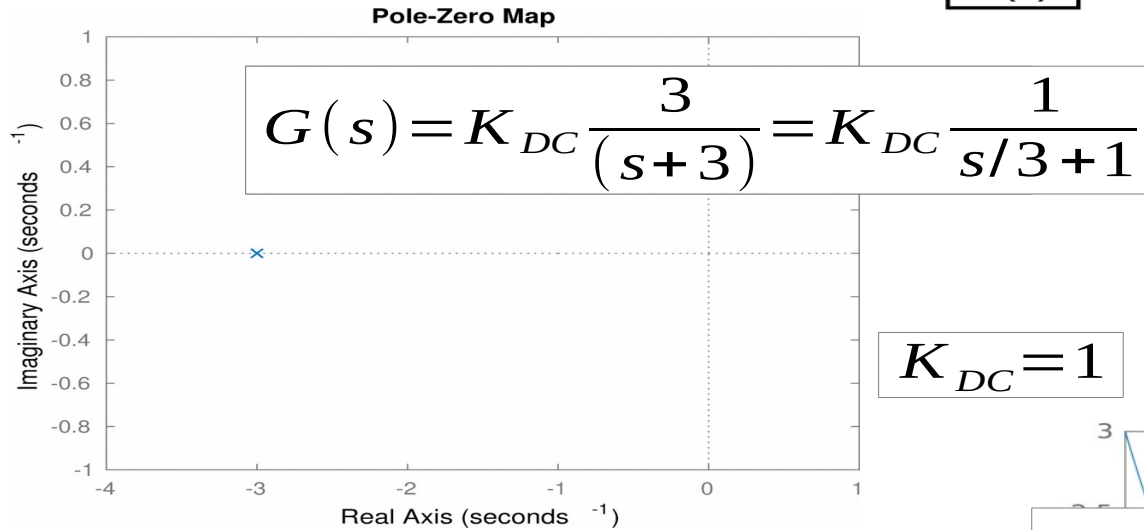
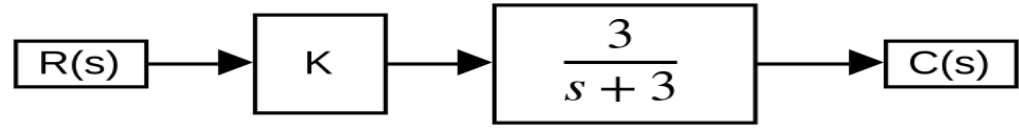


- Write the TF
- Evaluate stability
- Sketch the impulse response
- Sketch the step response
- Find the time-constant

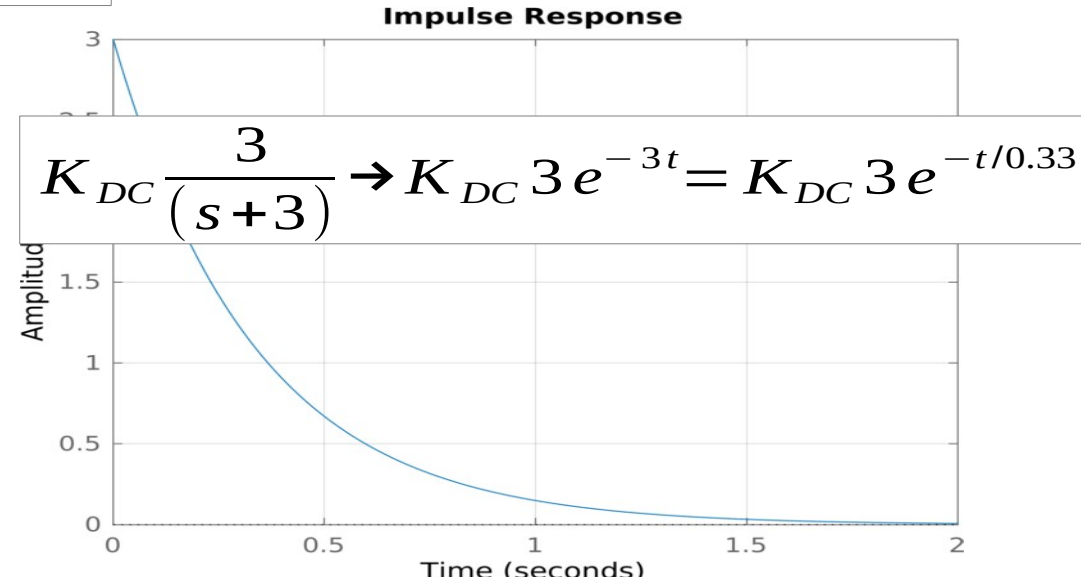
Thinking in the s-domain



Given this picture

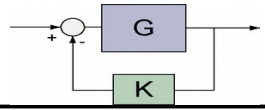


$$K_{DC} = 1$$

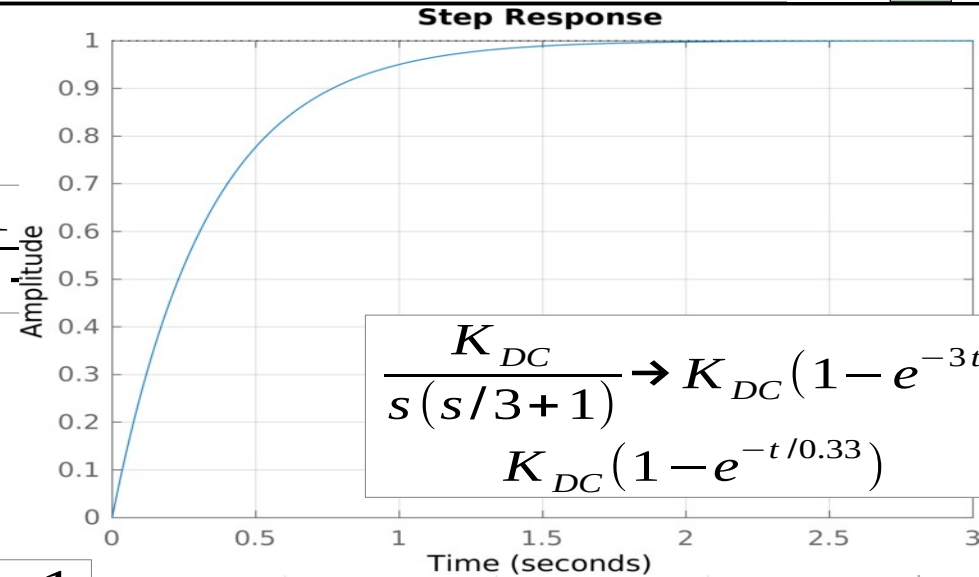
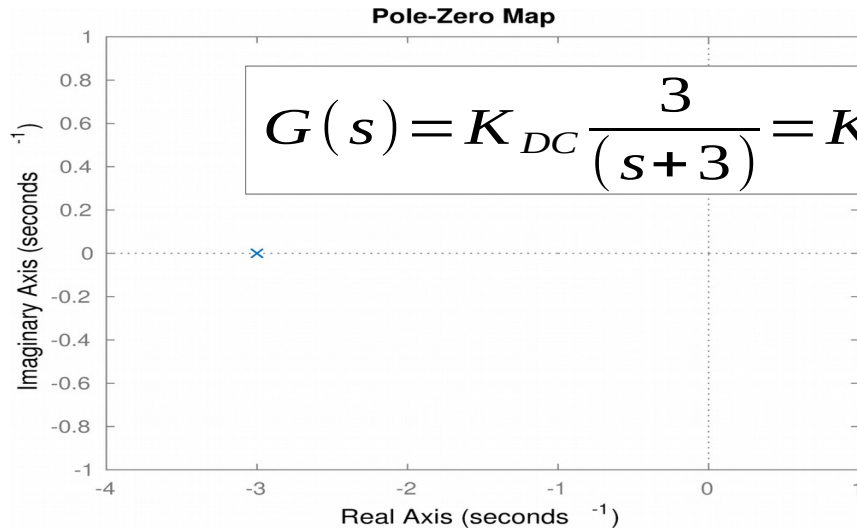


- Write the TF
- Evaluate stability
- Sketch the impulse response
- Sketch the step response
- Find the time-constant

Thinking in the s-domain



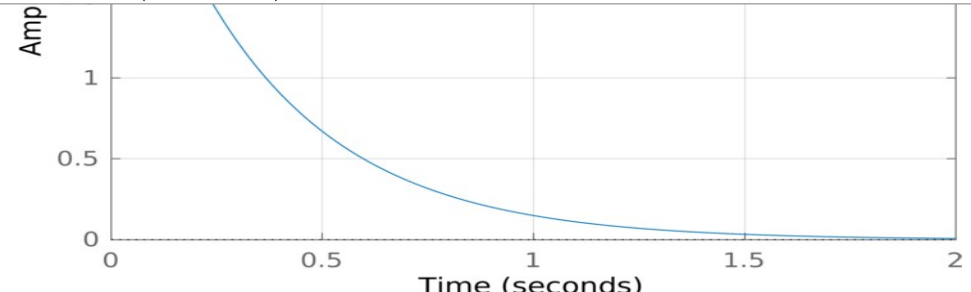
Given this picture

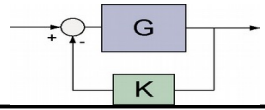


$$K_{DC} = 1$$

- Write the TF
- Evaluate stability
- Sketch the impulse response
- **Sketch the step response**
- Find the time-constant

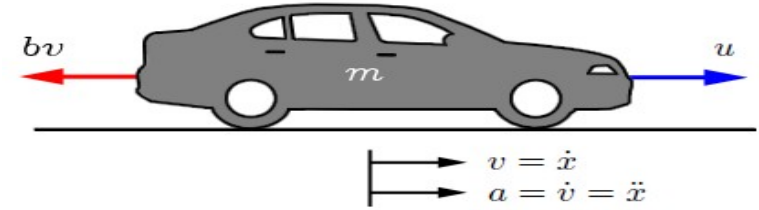
$$K_{DC} \frac{3}{(s+3)} \rightarrow K_{DC} 3e^{-3t} = K_{DC} 3e^{-t/0.33}$$





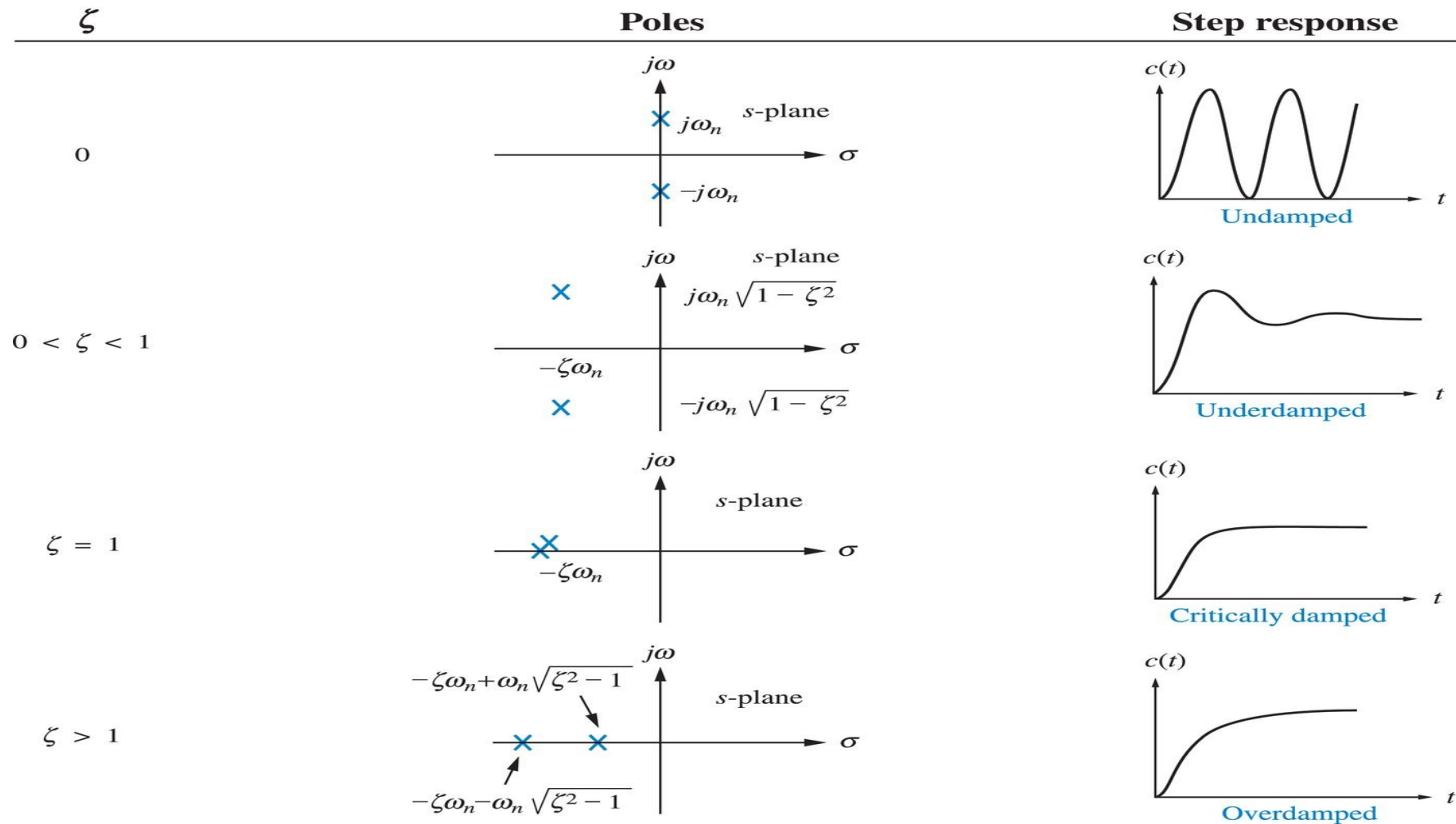
Simple Car Model

- By inspection
 - DC Gain =
 - Time constant =
- Sketch
 - p-z map
 - Step response

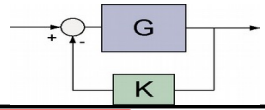


$$G(s) = \frac{1/1000}{(s + 50/1000)} = \frac{1/50}{(20)s + 1}$$

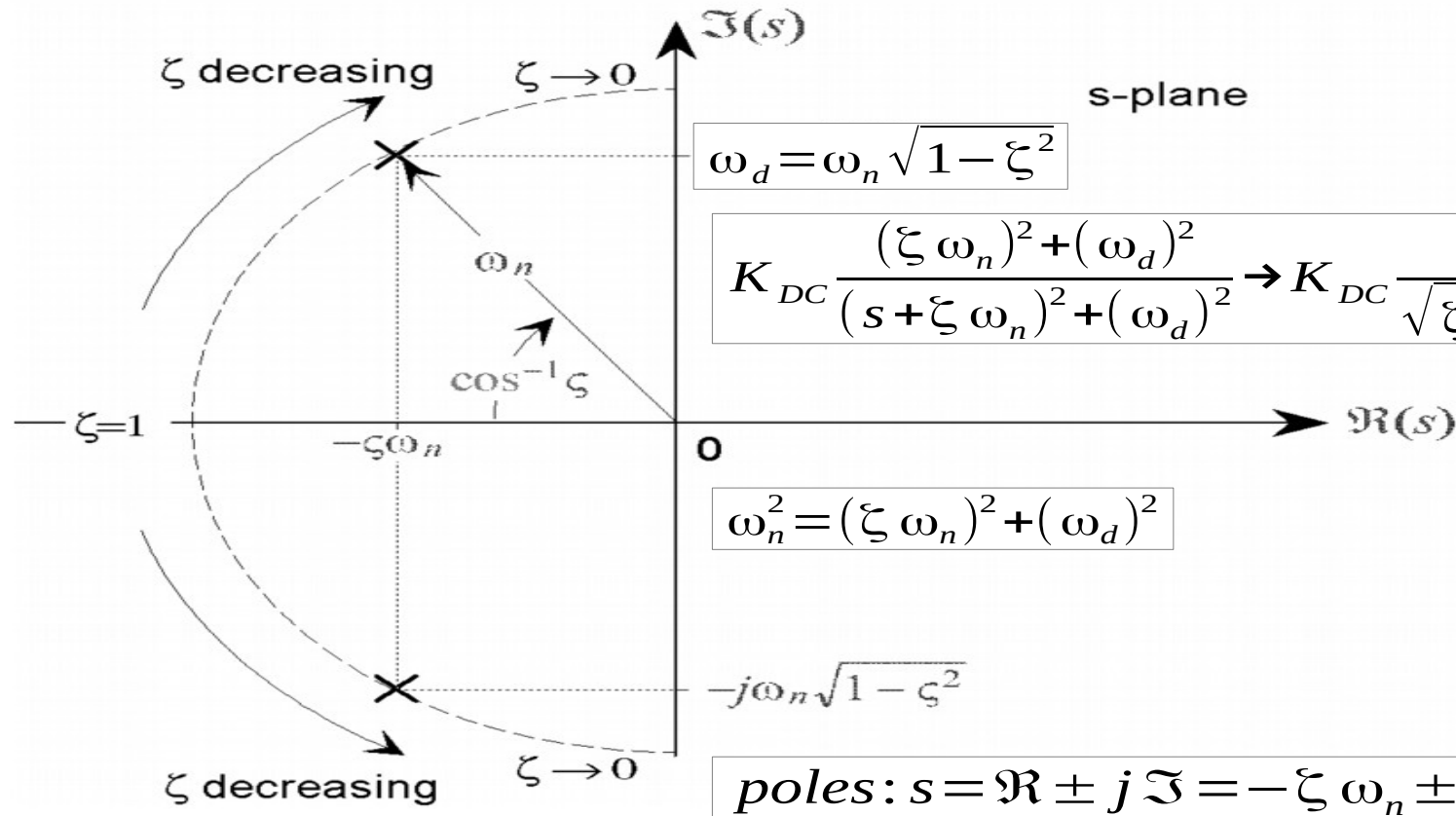
$$K \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



Underdamped: Complex Pole Geometry



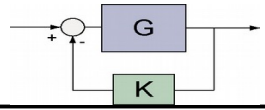
$$K_{DC} \frac{\Re^2 + \Im^2}{(s + \Re)^2 + (\Im)^2} = K_{DC} \frac{(\zeta \omega_n)^2 + (\omega_d)^2}{(s + \zeta \omega_n)^2 + (\omega_d)^2} = K_{DC} \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$



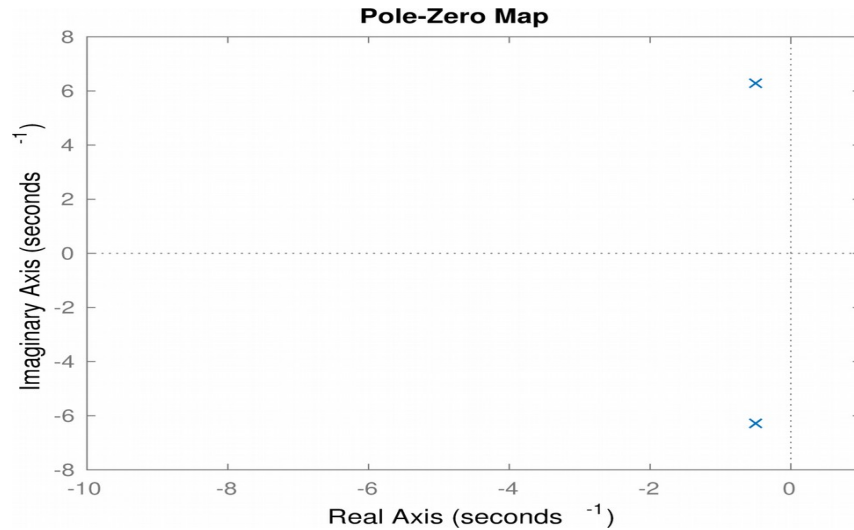
$$K_{DC} \frac{(\zeta \omega_n)^2 + (\omega_d)^2}{(s + \zeta \omega_n)^2 + (\omega_d)^2} \rightarrow K_{DC} \frac{\omega_n}{\sqrt{\zeta^2 - 1}} (e^{-\zeta \omega_n t}) \sin(\omega_d t)$$

$$\omega_n^2 = (\zeta \omega_n)^2 + (\omega_d)^2$$

$$\text{poles: } s = \Re \pm j \Im = -\zeta \omega_n \pm j \omega_d$$

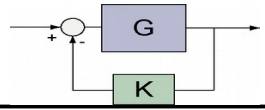


Given this picture

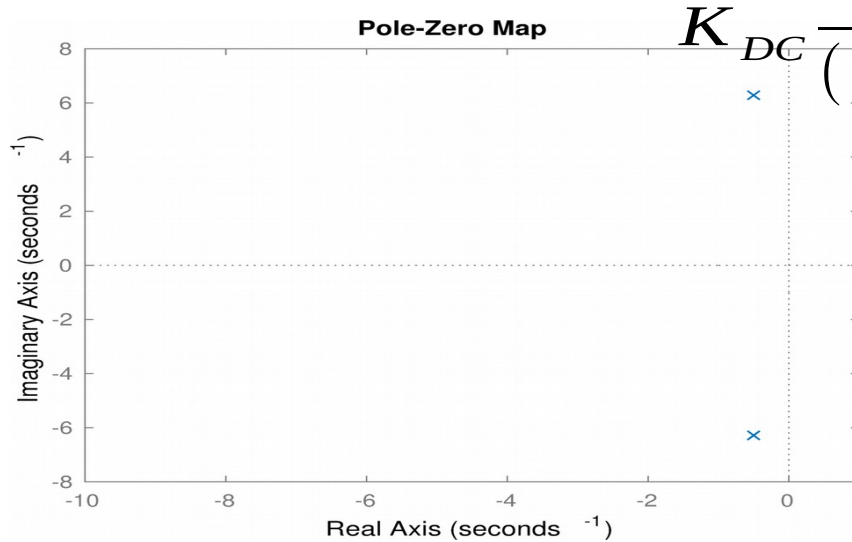


- Write the TF
- Evaluate stability
- Sketch the impulse response
- Sketch the step response

Thinking in the s-domain

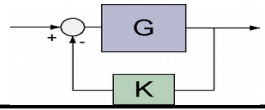


Given this picture

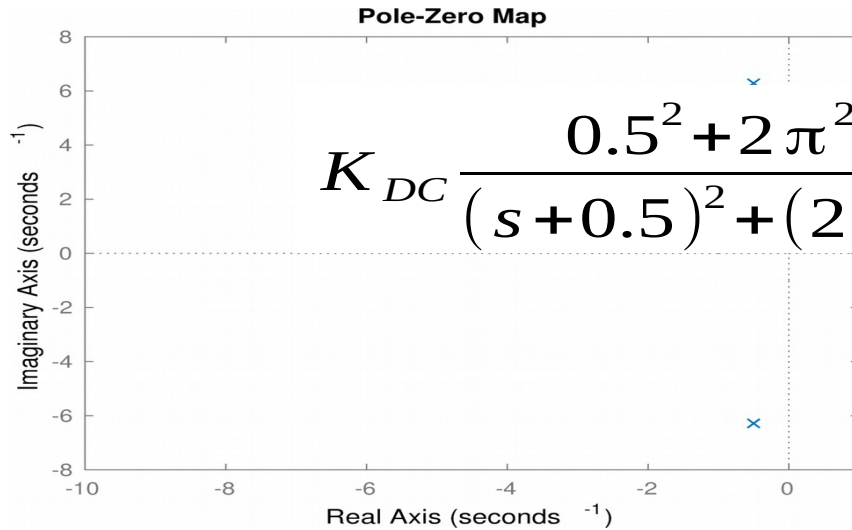


$$K_{DC} \frac{\omega_n^2}{(s + \zeta \omega_n)^2 + (\omega_d)^2} \rightarrow K_{DC} \frac{0.5^2 + 2\pi^2}{(s + 0.5)^2 + (2\pi)^2}$$

- Write the TF
- Evaluate stability
- Sketch the impulse response
- Sketch the step response

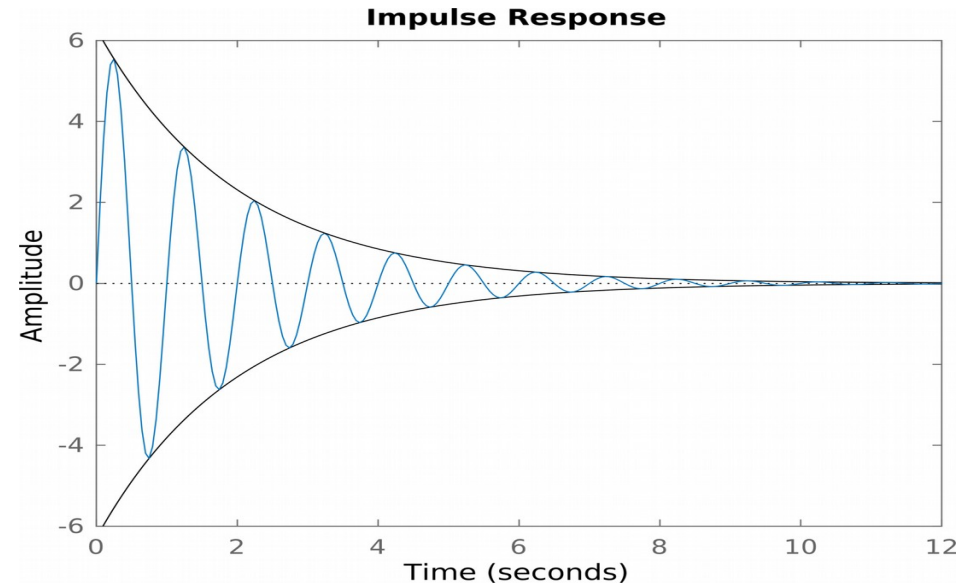


Given this picture

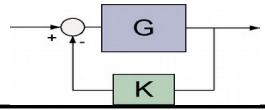


$$K_{DC} \frac{0.5^2 + 2\pi^2}{(s + 0.5)^2 + (2\pi)^2} \rightarrow K_{DC} \frac{\sqrt{(0.5^2 + 2\pi^2)}}{2\pi} (e^{-0.5t}) \sin(2\pi t)$$

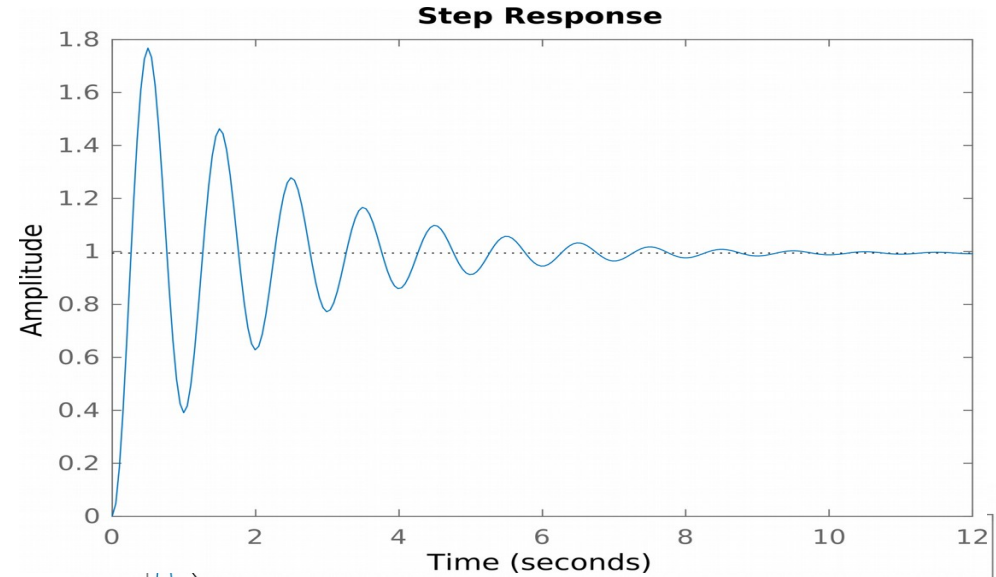
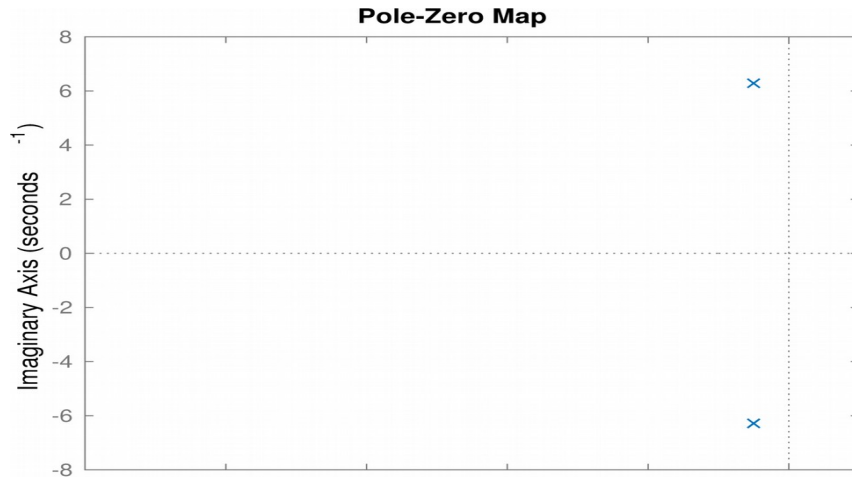
- Write the TF
- Evaluate stability
- **Sketch the impulse response**
- Sketch the step response



Thinking in the s-domain



Given this picture

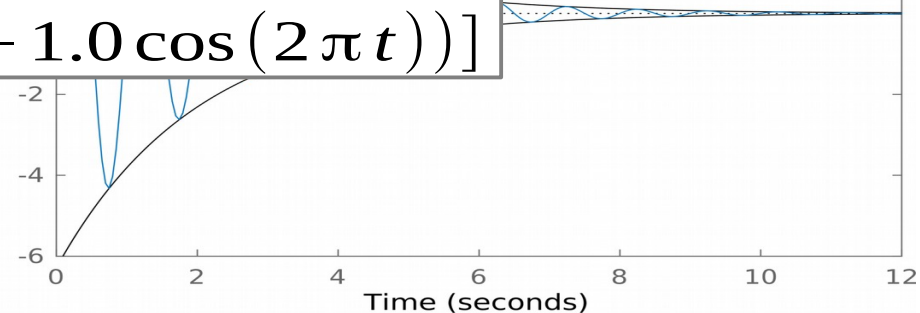


$$C(s) = \frac{1}{s} K \frac{2\pi^2}{(s+0.5)^2 + (2\pi)^2} \rightarrow$$

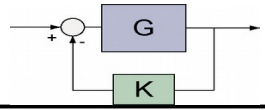
$$K [1.0 \mu(t) + e^{-0.5t} (0.08 \sin(2\pi t) - 1.0 \cos(2\pi t))] +$$

• Sketch the impulse response

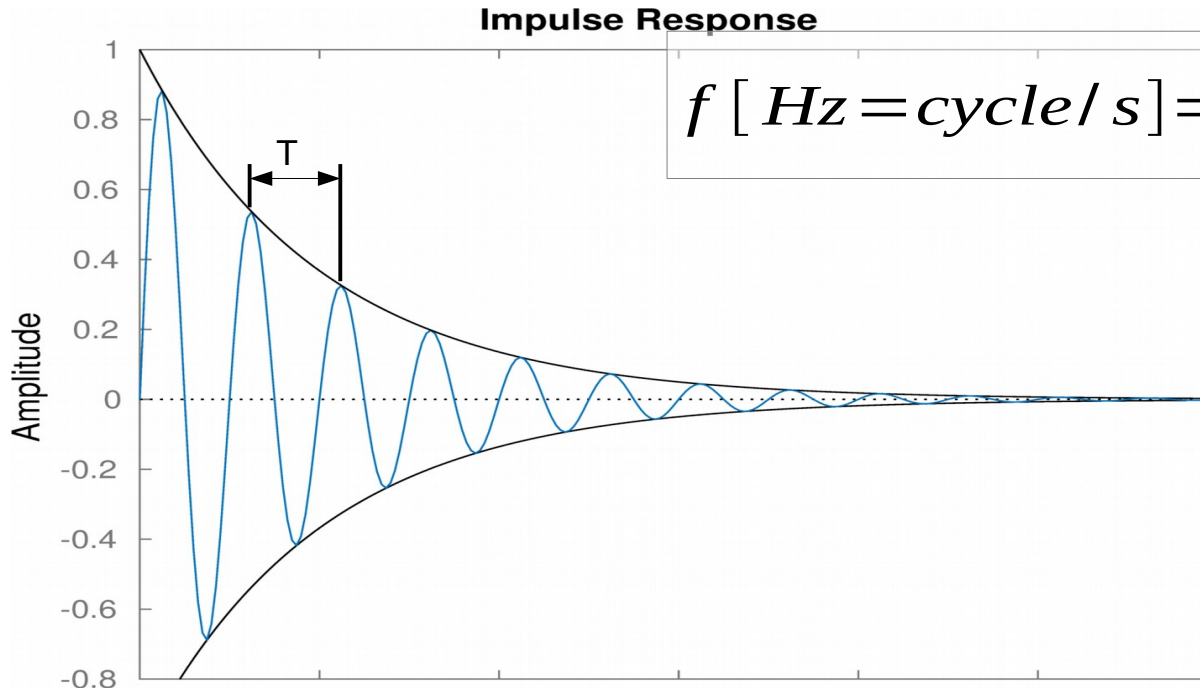
• Sketch the step response



Second Order Response

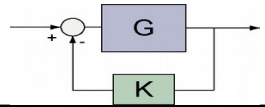


$$T[s] = \frac{1}{\omega_d [\text{rad/s}]}$$



$$f [\text{Hz} = \text{cycle/s}] = \frac{\omega_d [\text{rad/s}]}{2\pi [\text{rad/cycle}]}$$

$$K_{DC} \frac{0.5^2 + 2\pi^2}{(s + 0.5)^2 + (2\pi)^2} \rightarrow K_{DC} \frac{\sqrt{(0.5^2 + 2\pi^2)}}{2\pi} (e^{-0.5t}) \sin(2\pi t)$$



A system is...

Stable if the natural response approaches zero as time approaches infinity

- TEST: Are **all** poles in the LHP?
(negative real component)

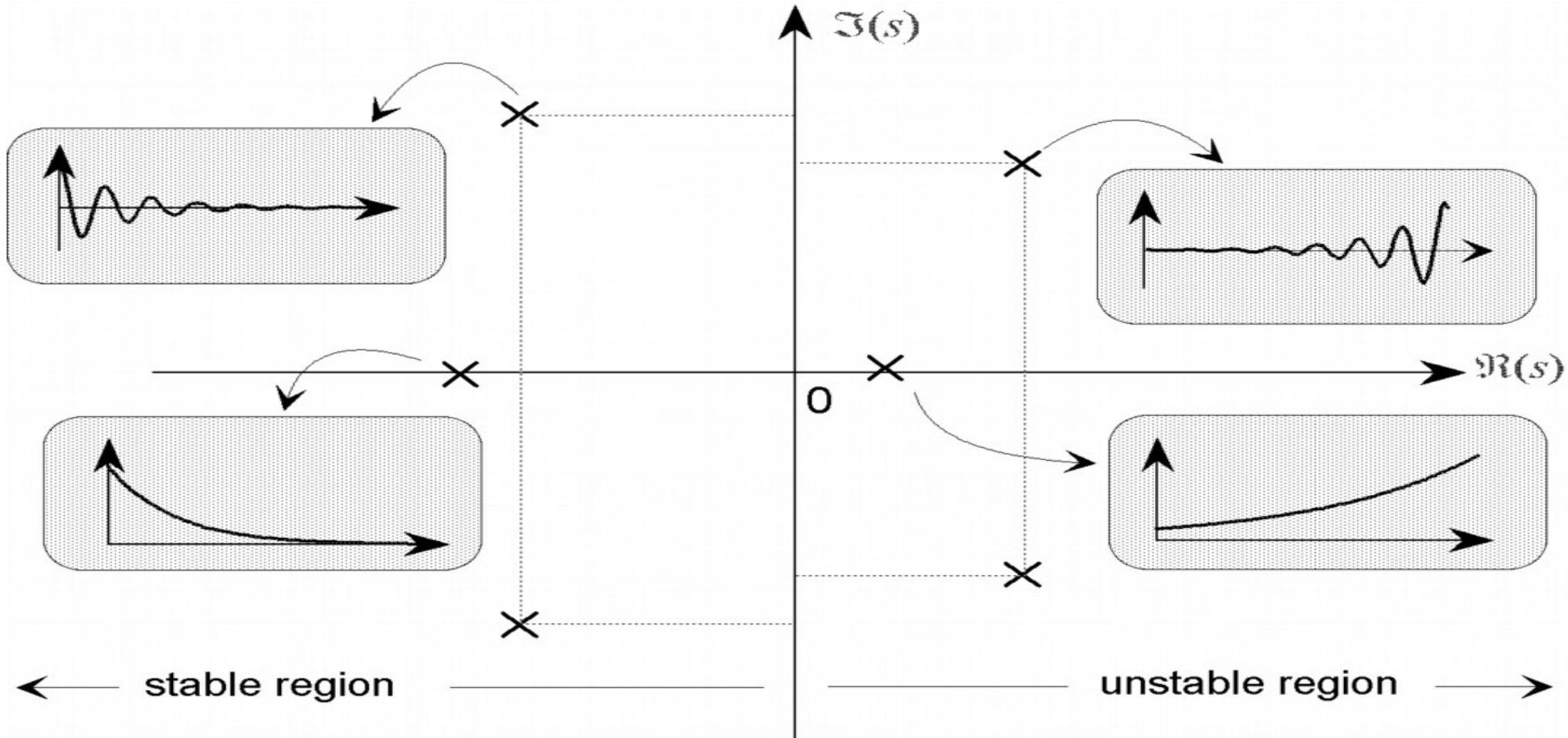
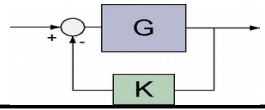
Unstable if the natural response grows without bound as time approaches infinity

- TEST: Are **any** poles in the RHP?
(positive real component)

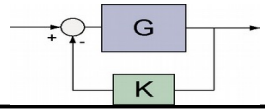
Marginally stable if the natural response neither decays nor grows but remains constant or oscillates

- TEST: Are any poles on the imaginary axis
(real component is zero)

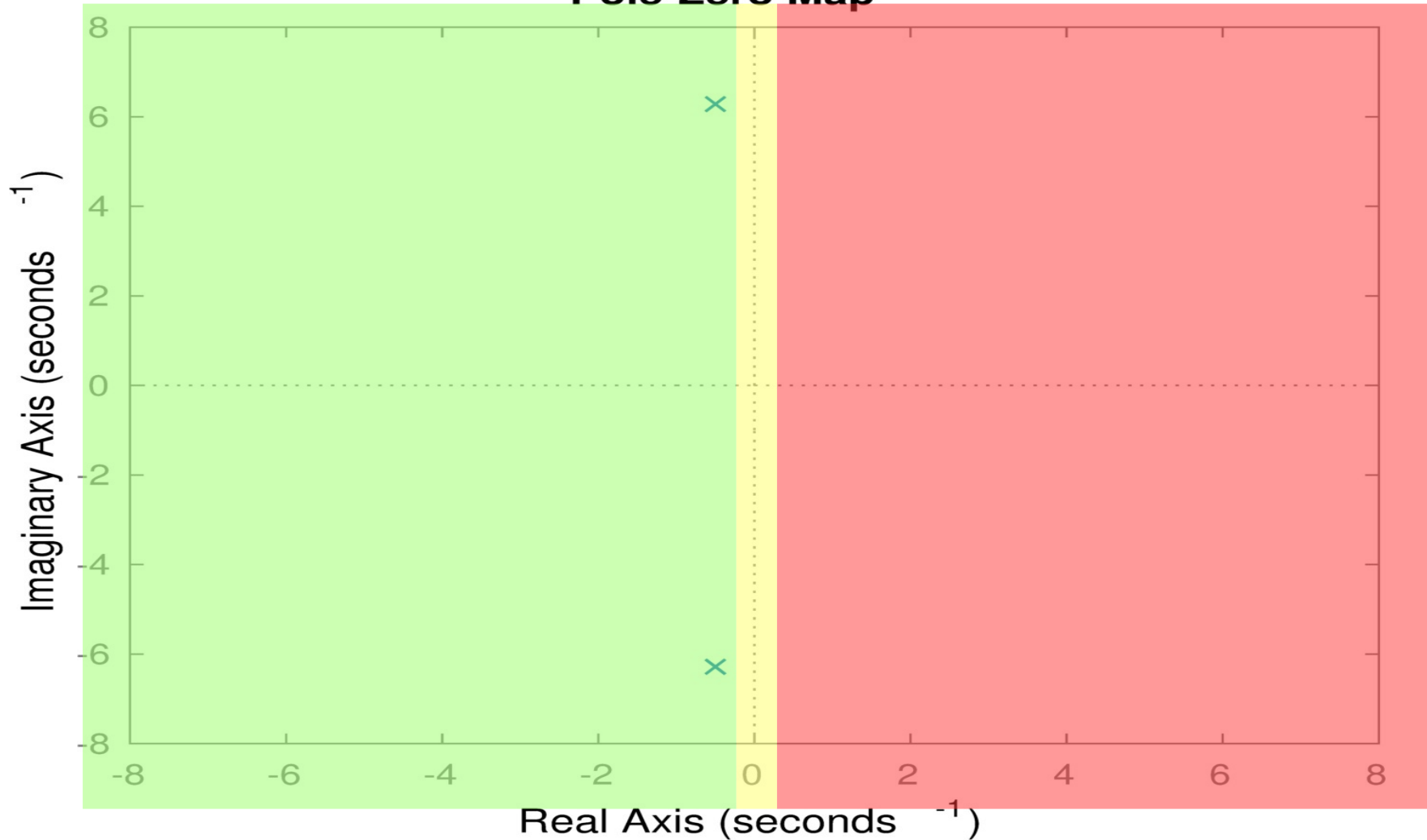
General Ideas: Stable, Real, Complex



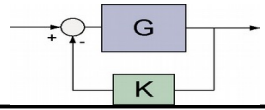
Simple Stability



Pole-Zero Map



Performance Specs: First Order

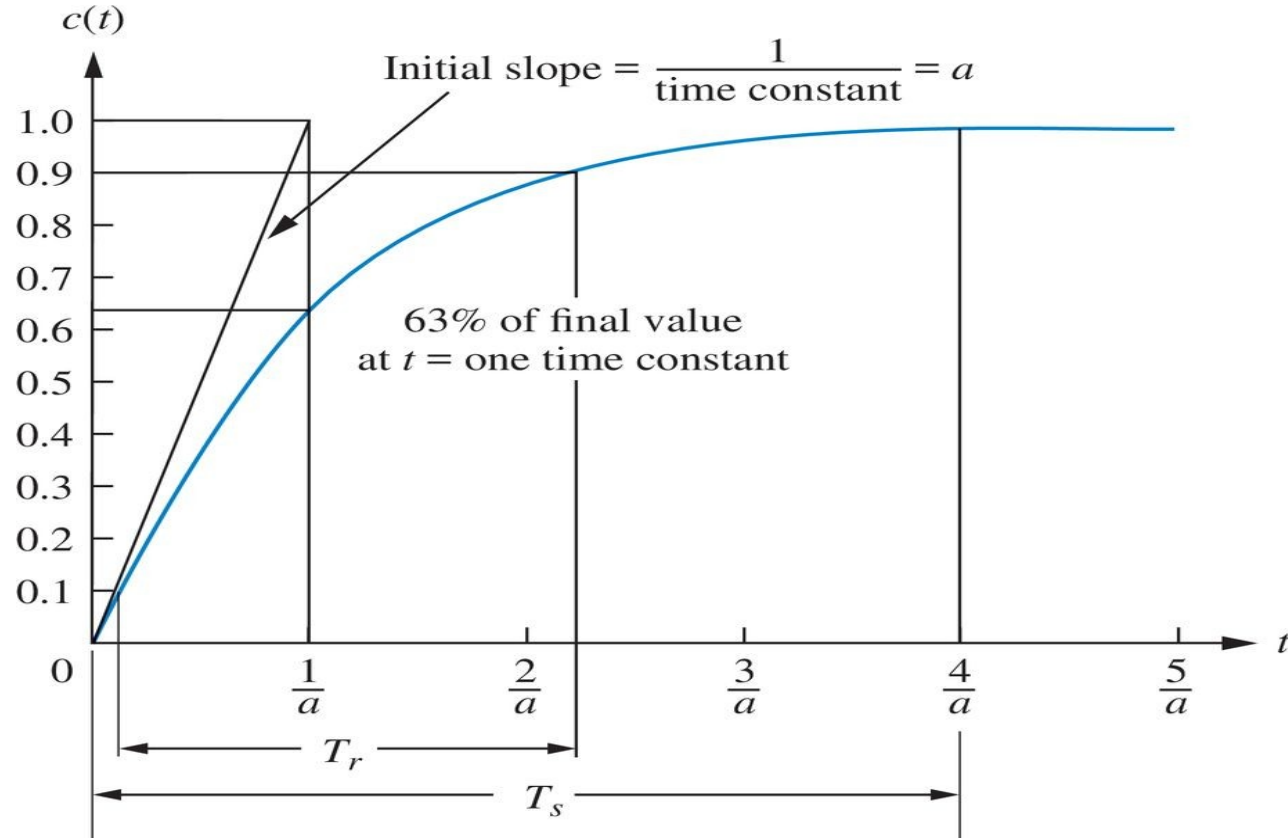


$$G(s) = K \frac{a}{s+a} = K \frac{1}{\tau s + 1}$$

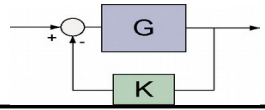
$$T_c = \tau = 1/a$$

$$T_r = \frac{2.2}{a} = 2.2(T_c)$$

$$T_s(98\%) = \frac{4}{a} = 4(T_c)$$



Perf. Specs: Second Order



$$K \frac{(\zeta \omega_n)^2 + (\omega_d)^2}{(s + \zeta \omega_n)^2 + (\omega_d)^2} = K \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

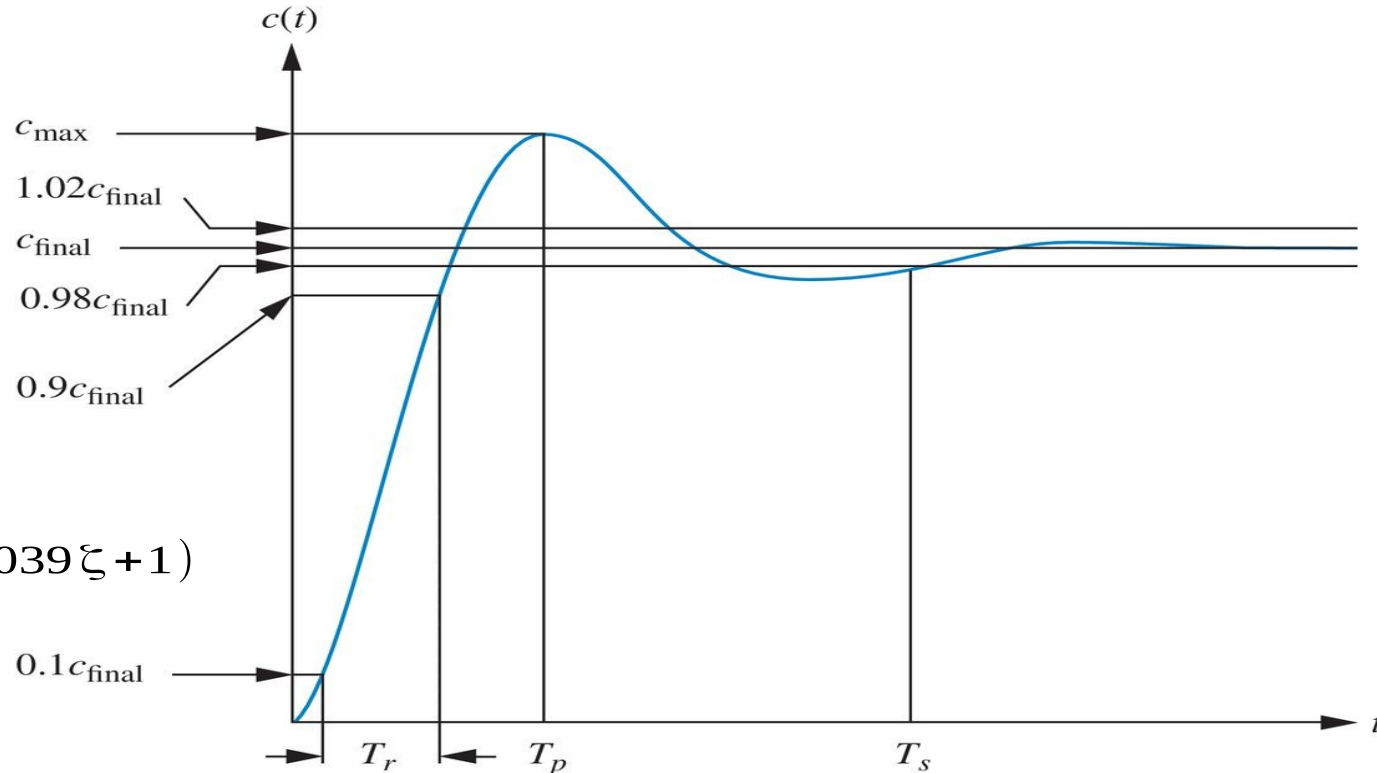
$$\%OS = e^{-(\zeta \pi / \sqrt{1 - \zeta^2})} \times 100$$

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$

$$T_s = \frac{4}{\zeta \omega_n}$$

$$T_r = \frac{1}{\omega_n} (1.76 \zeta^3 - 0.417 \zeta^2 + 1.039 \zeta + 1)$$

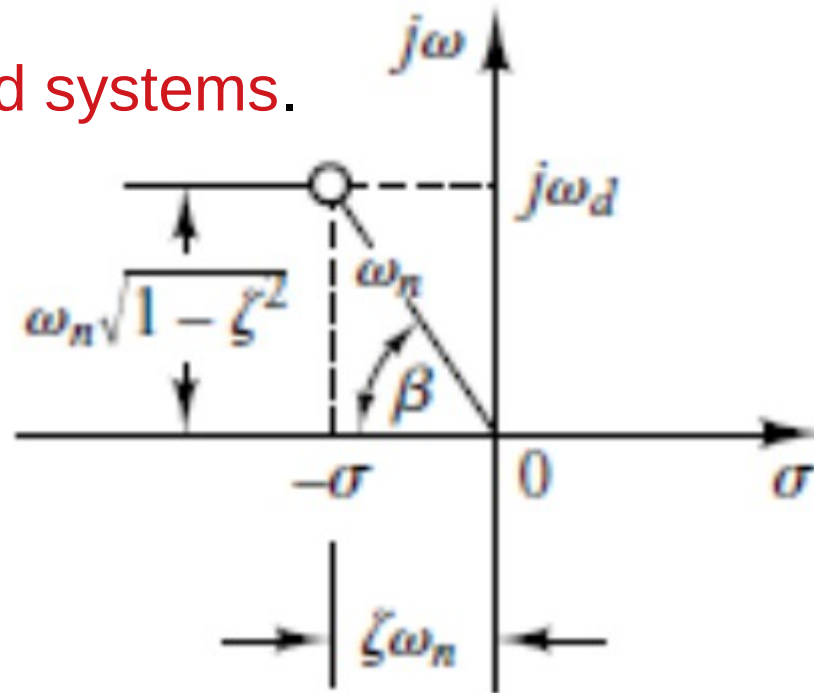
$0 < \zeta < 0.9$

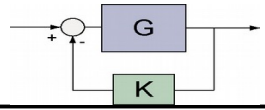


For the 0-100% rise time.

Only applicable to underdamped systems.

$$T_{r2} = \frac{\pi - \beta}{\omega_d}$$





Numerical Evaluation

$$\text{Normalized rise time} = \omega_n T_r$$

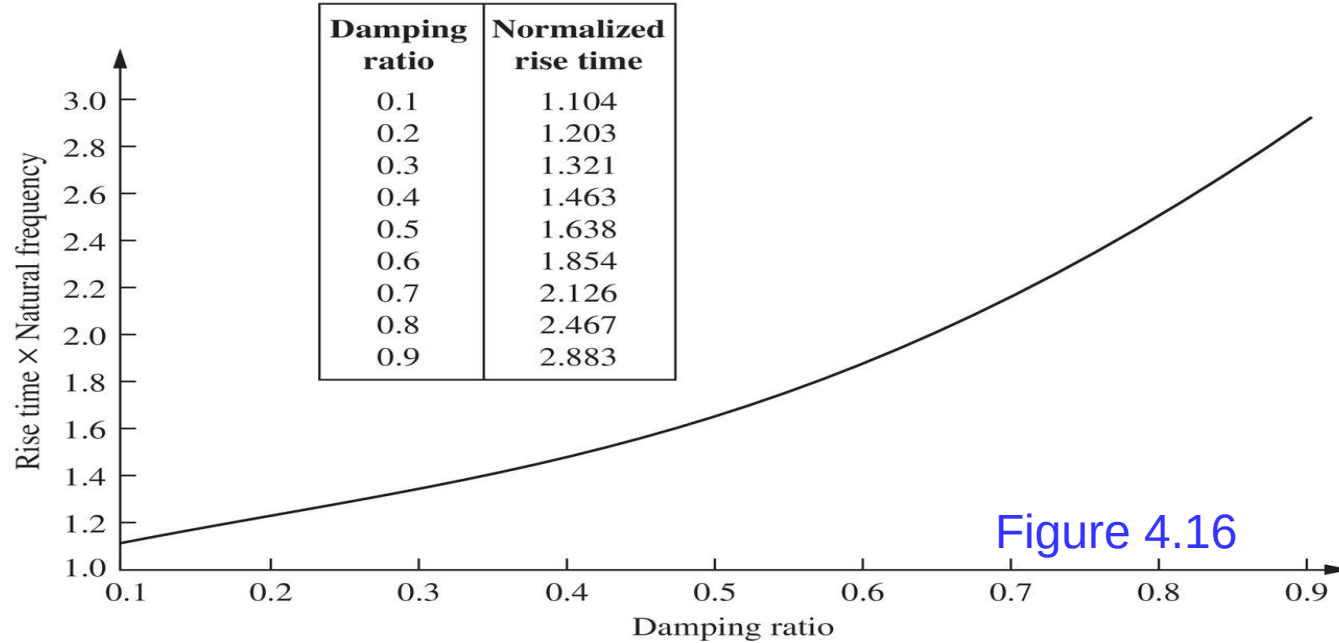
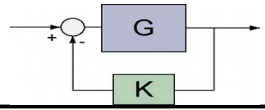


Figure 4.16

Given: ζ and ω_n ; Find rise time

- For given damping ratio, use graph to lookup: $\omega_n T_r$
- Solve for:
$$T_r = \frac{(\omega_n T_r)}{\omega_n}$$



Nise 4.2 and 4.3

- Find step output for first-order systems
- Time constant, rise time and settling time
- Plot step response with MATLAB (step()) and annotate to indicate the time constant.

Nise 4.8

- Plot poles and zeros by hand
- Write “general form” of step response
- State type of response: overdamped, underdamped, etc.

$$c(t) = K[1 - e^{-at}]$$

$$c(t) = K[1 - Ae^{-\sigma_1 t} - Be^{-\sigma_2 t}]$$

$$c(t) = K[1 - Ae^{-\sigma_d t} \cos(\omega_d t - \phi)]$$

$$c(t) = K[1 - A \cos(\omega_1 t - \phi)]$$

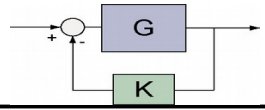
$$c(t) = K[1 - Ae^{-\sigma_1 t} - Bte^{-\sigma_1 t}]$$

$$c(t) = A + Be^{-\sigma_1 t} + Ce^{-\sigma_2 t}$$

$$c(t) = A + Be^{-\sigma_d t} \cos(\omega_d t + \phi)$$

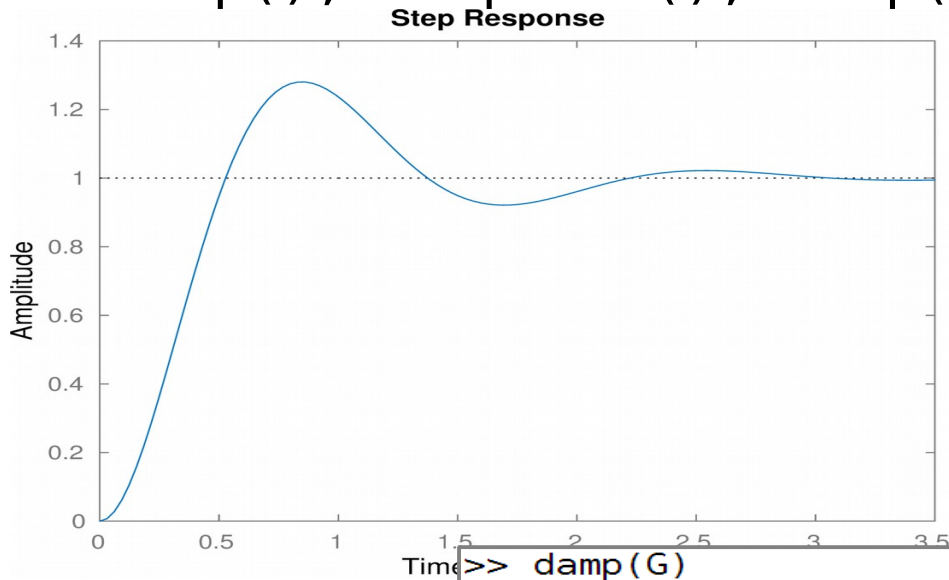
$$c(t) = A + B \cos(\omega_1 t + \phi)$$

$$c(t) = A + Be^{-\sigma_1 t} + Cte^{-\sigma_1 t}$$



Nise 4.20 and 4.21

- Find second-order parameters and metrics
- Three views: by hand, graph and MATLAB `step()`, `stepinfo()`, `damp()`



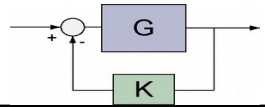
```
>> stepinfo(G)
ans =
    RiseTime: 0.3569
    SettlingTime: 2.6570
    SettlingMin: 0.9213
    SettlingMax: 1.2803
    Overshoot: 28.0255
    Undershoot: 0
           Peak: 1.2803
    PeakTime: 0.8596
```

```
>> damp(G)
```

Pole	Damping	Frequency (rad/seconds)	Time Constant (seconds)
-1.50e+00 + 3.71e+00i	3.75e-01	4.00e+00	6.67e-01
-1.50e+00 - 3.71e+00i	3.75e-01	4.00e+00	6.67e-01

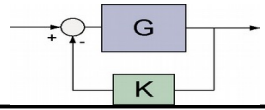


HW3



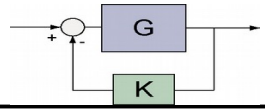
Nise 4.23

- Given performance metrics (%OS, T_p , T_s)
Find pole locations (damping ratio and natural frequency)

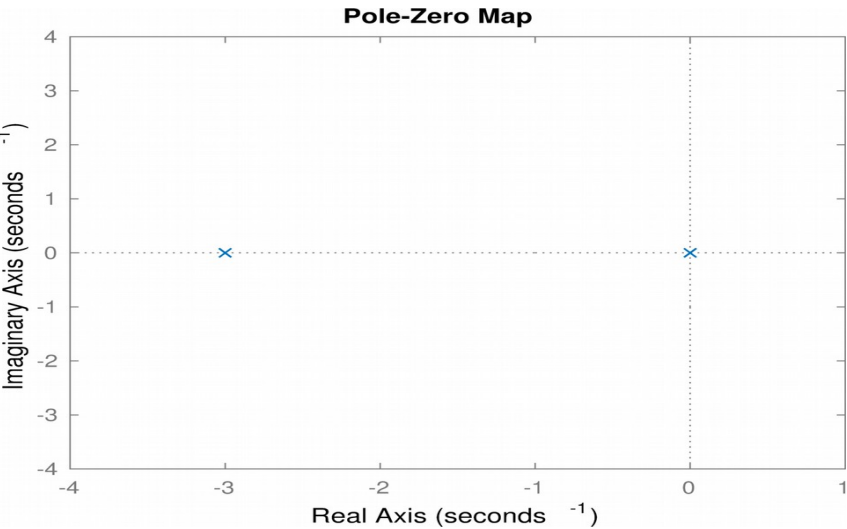


More Examples

Thinking in the s-domain

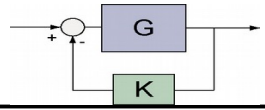


Given this picture

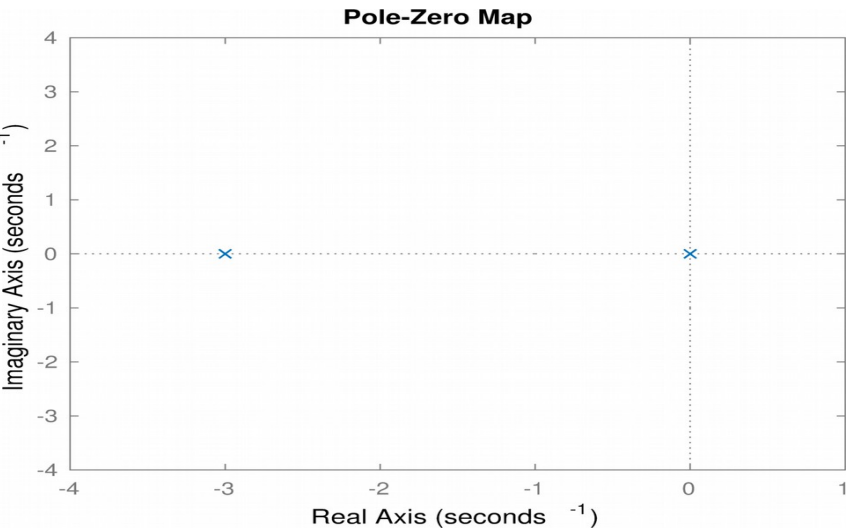


- Write the TF
- Sketch the step response
- Sketch the impulse response
- Find the time-constant
- Evaluate stability

Thinking in the s-domain



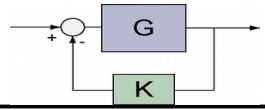
Given this picture



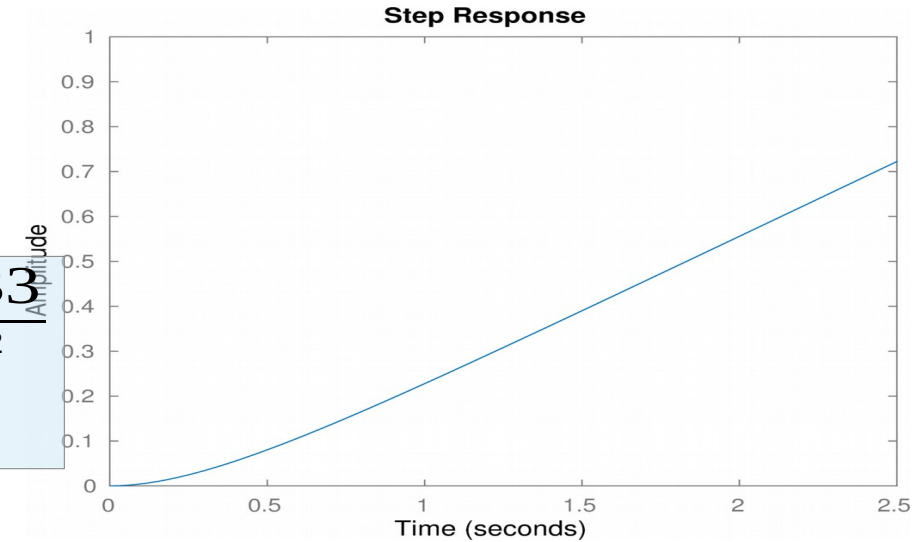
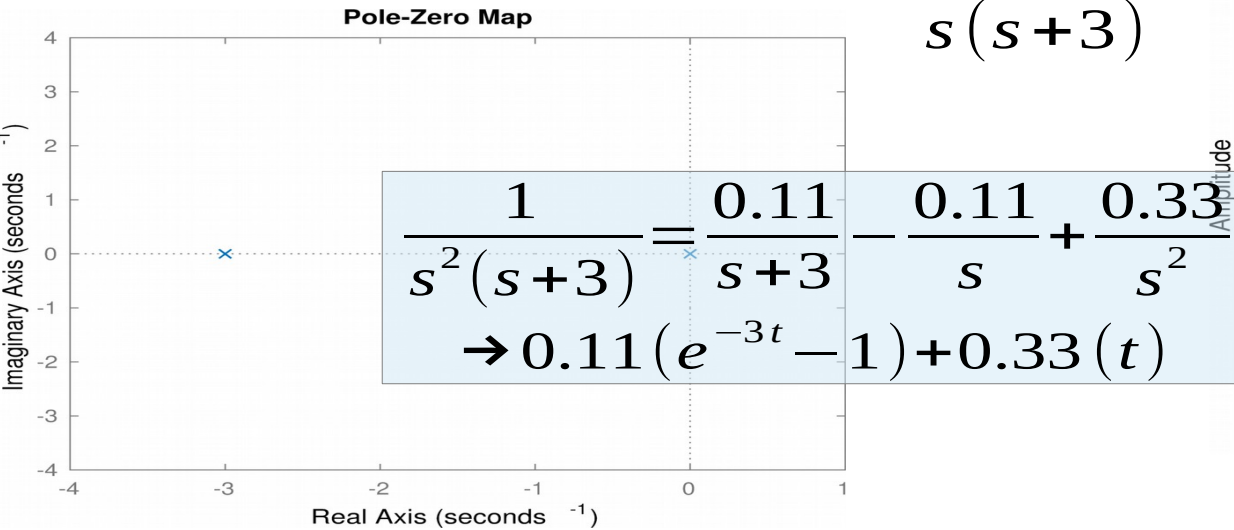
$$\frac{1}{s(s+3)}$$

- Write the TF
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- Sketch the impulse response
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- Evaluate stability

Thinking in the s-domain

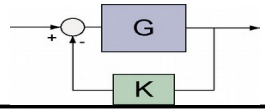


Given this picture



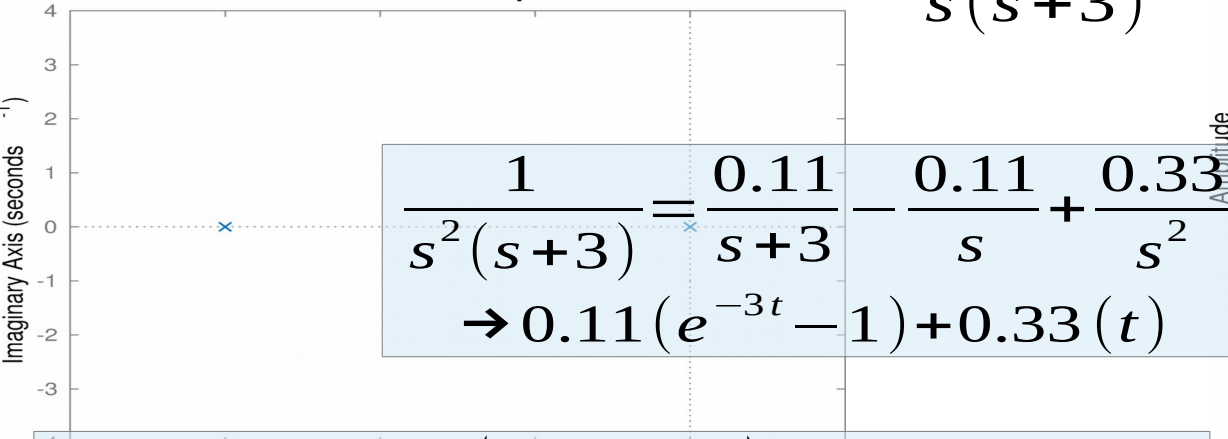
- Write the TF
- Sketch the step response
- Sketch the impulse response
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- Evaluate stability

Thinking in the s-domain



Given this picture

Pole-Zero Map



$$\frac{1}{s^2(s+3)} = \frac{0.11}{s+3} - \frac{0.11}{s} + \frac{0.33}{s^2}$$

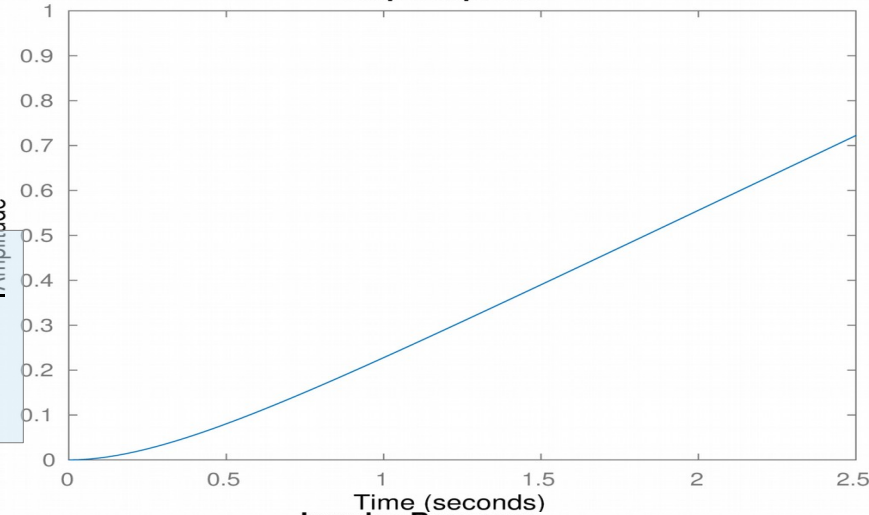
$$\rightarrow 0.11(e^{-3t} - 1) + 0.33(t)$$

$$\frac{1}{s(s+3)}$$

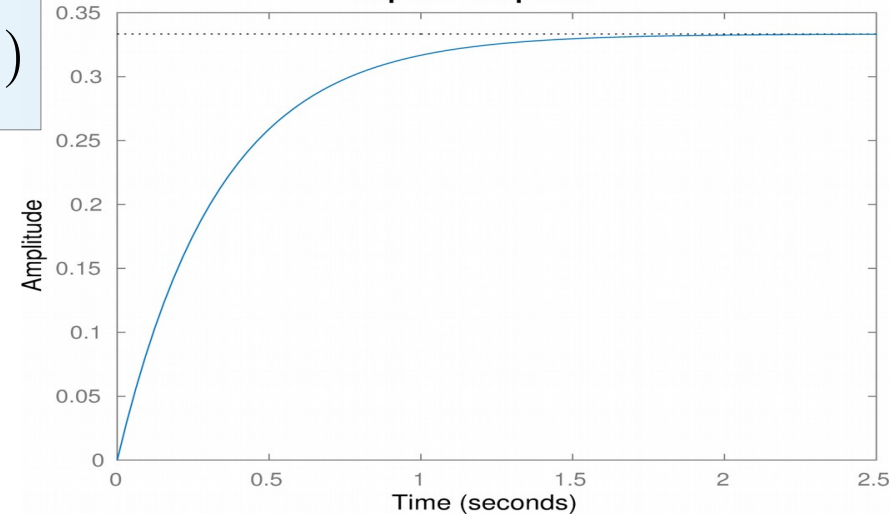
$$\frac{1}{s(s+3)} = 0.33 \left(\frac{1}{s} - \frac{1}{s+3} \right) \rightarrow 0.33(1 - e^{-3t})$$

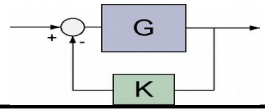
- Write the TF
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Step Response

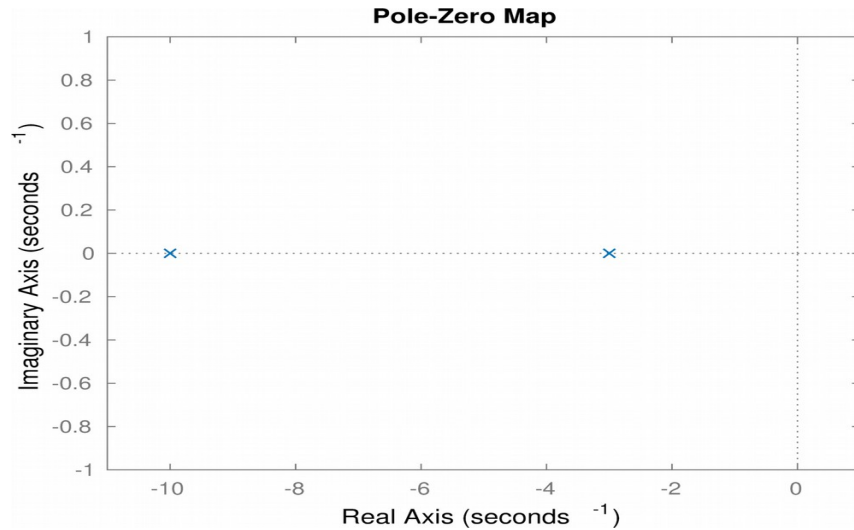


Impulse Response



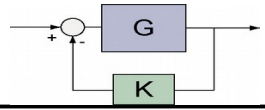


Given this picture



- Write the TF
- Sketch the step response
- Sketch the impulse response
- Find the time-constant
- Evaluate stability

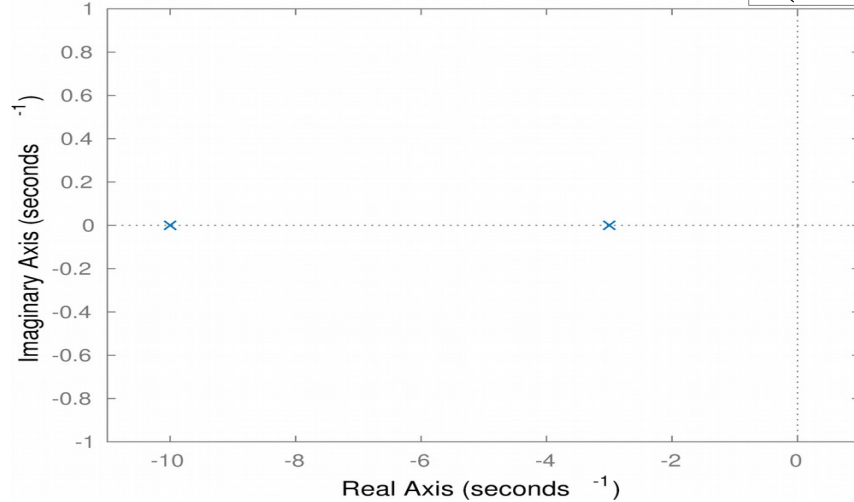
Thinking in the s-domain



Given this picture

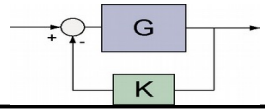
$$\frac{100}{(s+3)(s+10)}$$

Pole-Zero Map

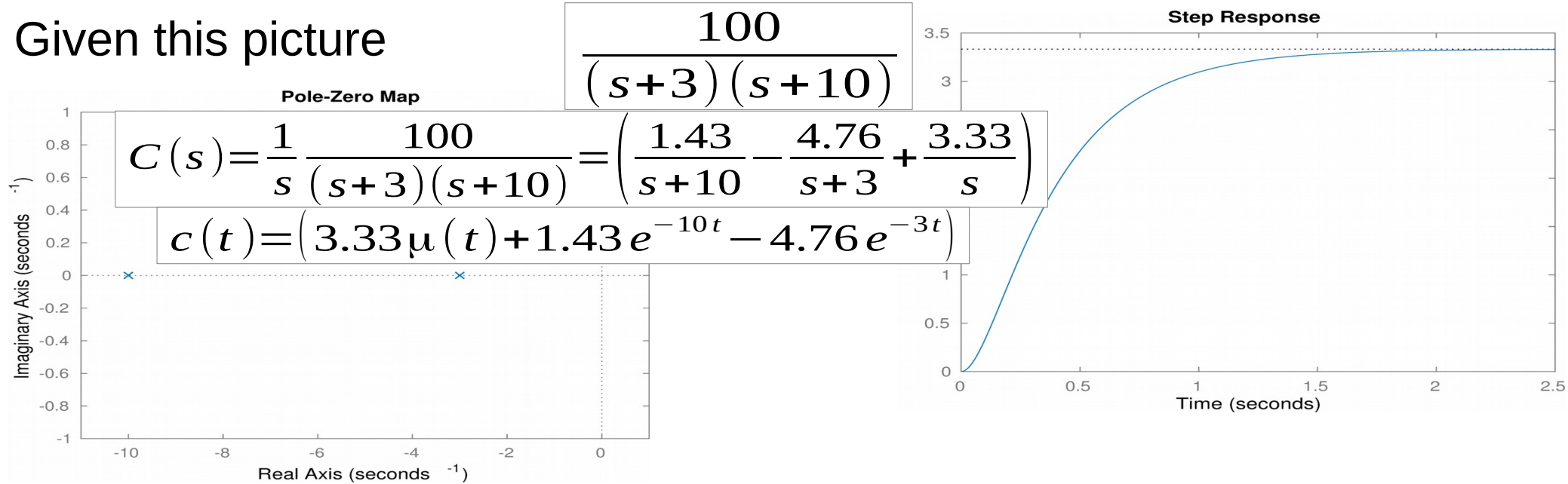


- Write the TF
- Sketch the step response
- Evaluate stability

Thinking in the s-domain



Given this picture



- Write the TF
- Sketch the step response
- Evaluate stability